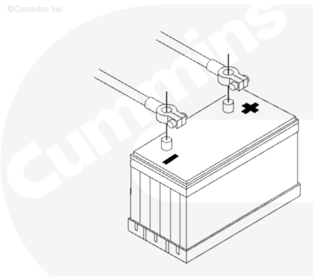
Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

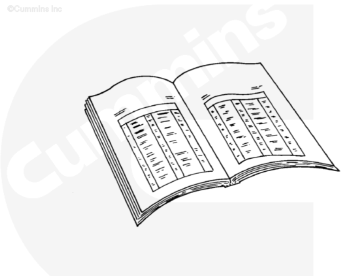


13900050

- Remove the batteries.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



ck800wa

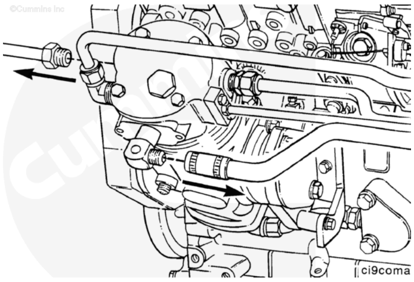
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

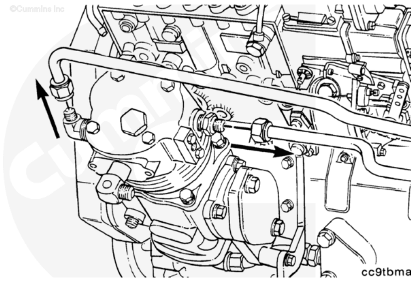
- Steam-clean the air compressor and dry with compressed air.
- Drain the engine coolant if the air compressor has a liquid cooled cylinder head. If compressor is air cooled, then the engine coolant need **not** be drained. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html) if coolant needs to be drained.
- Open the draincock on the wet tank to release air from the system. Close the draincock after the pressure is released.

Remove

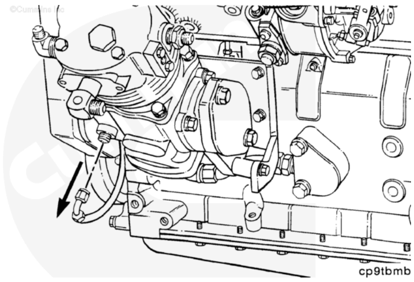
Remove the air connections from the air compressor.



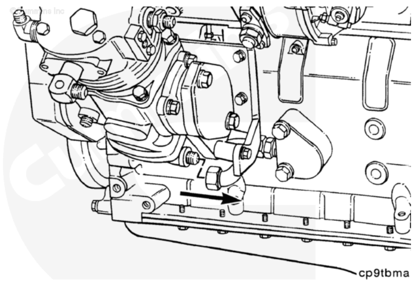
Remove the coolant lines from the air compressor (does **not** apply to air cooled compressors).



Remove the oil supply line.



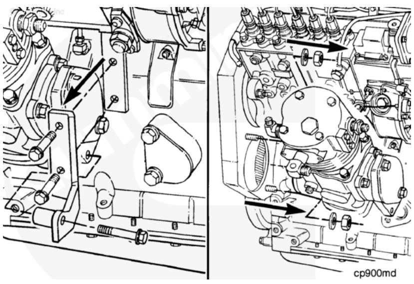
Remove the oil return line from the bottom of the air compressor.



Remove the air compressor support bracket and capscrews.

Remove the air compressor mounting nuts.

Remove the air compressor.

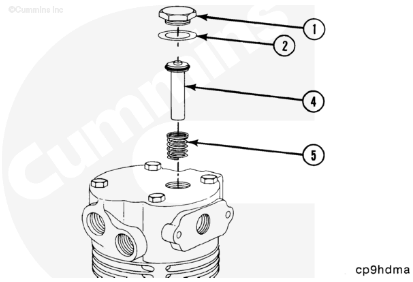


Disassemble

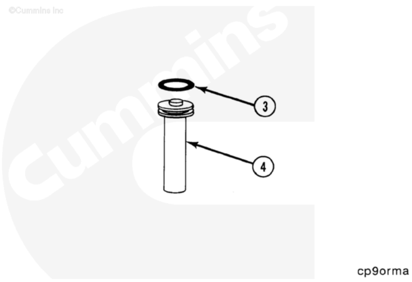
Remove the following parts:

- Unloader cover (1)
- Copper washer (2)
- Unloader pin (4)
- Spring (5).

Discard the copper washer.



Remove and discard the o-ring (3) from the unloader pin (4).

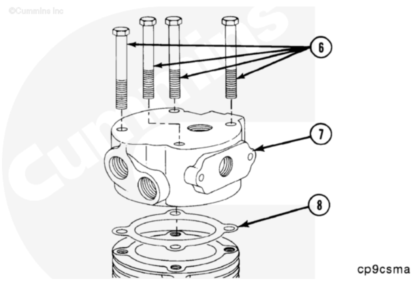


Remove the four cylinder head capscrews (6).

Remove the cylinder head (7).

Remove and discard the cylinder head gasket (8).

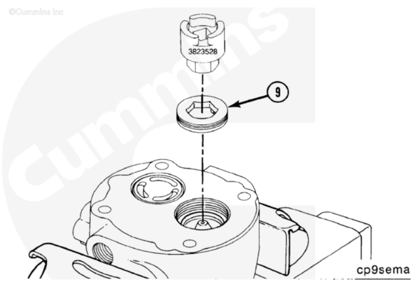
Service Tip: Scribe a mark to show proper head orientation before removing the head.



Air Compressor Seat Socket or 3/4-Inch Allen Wrench, Part Number 3823528

Install the head with the bottom side up in a soft-jawed vise.

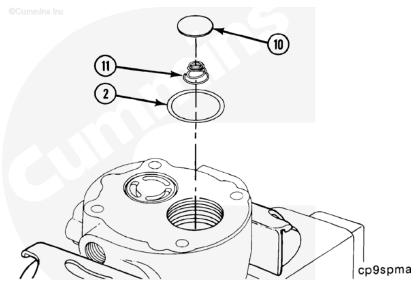
Remove the exhaust valve seat (9).



Remove the following parts:

- Exhaust valve disc (10)
- Spring (11)
- Copper washer (2).

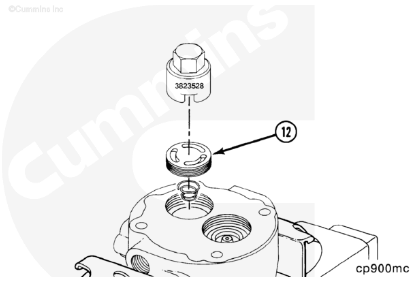
Discard the copper washer.



Note : The exhaust valve stop is pressed in place and must not be removed.

Air Compressor Seat Socket, Part Number 3823528

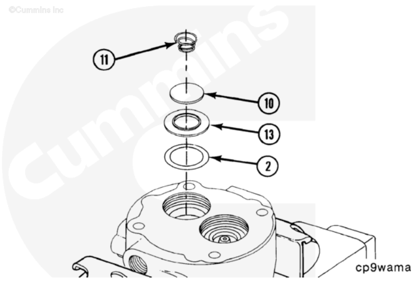
Remove the inlet valve cage (12).



Remove the following parts:

- Spring (11)
- Inlet valve disc (10)
- Inlet valve seat (13)
- Copper washer (2).

Discard the copper washer



Clean and Inspect for Reuse

⚠ WARNING ⚠

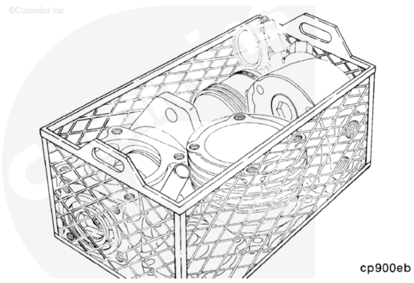
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Soak the parts in a kerosene emulsion-based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature, and soak time.

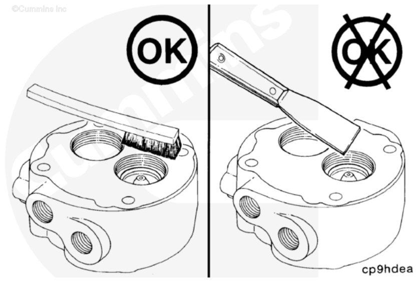
Dry with compressed air.



⚠ CAUTION ⚠

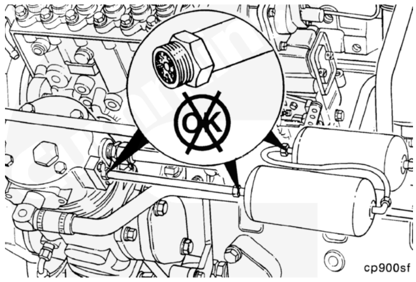
Do not use a scraper to remove carbon and scale. This can damage sealing surfaces.

Use a stiff, nonmetallic bristle brush to scrub the parts.



⚠ WARNING ⚠

The air discharge line must be capable of withstanding extreme heat and pressure to avoid the possibility of personal injury and property damage. Refer to the manufacturer's specifications.

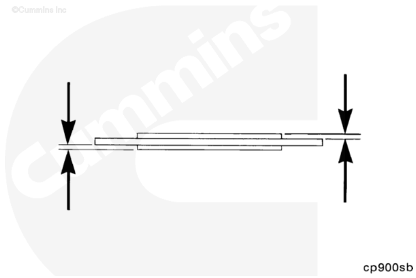


Continue to check for carbon buildup in the air discharge line connections up to the first or wet tank.

Clean or replace any lines and fittings with carbon deposits greater than 1.6 mm [0.06 (1/6) in]. Refer to the manufacturer's specifications for cleaning or replacement instructions.

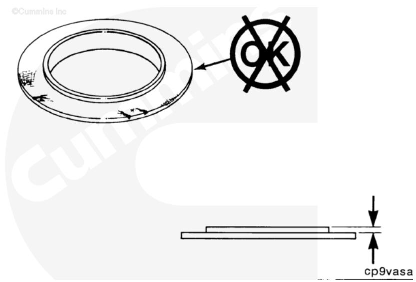
Valve Discs

Inspect and replace if cracked, pitted, or grooved in excess of 0.13 mm [0.005 in].



Inlet Valve Seat

Measure the distance from the valve seating surface to the surface that contacts the valve cage.

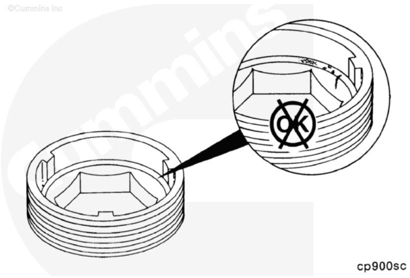


mm		in
0.597	MIN	0.0235
0.673	MAX	0.0265

Replace the intake valve seat if **not** within limits or if cracked or damaged.

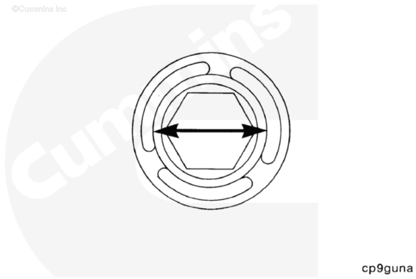
Exhaust Valve Seat

Inspect the seat for damage and wear.



Measure the valve guide diameter.

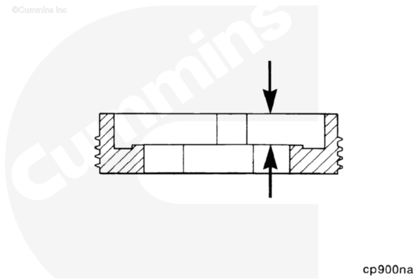
mm		in
25.53	MIN	1.005
25.65	MAX	1.010



Replace the exhaust valve seat if **not** within limits.

Measure the distance from the top of the valve seat to the valve seating surface.

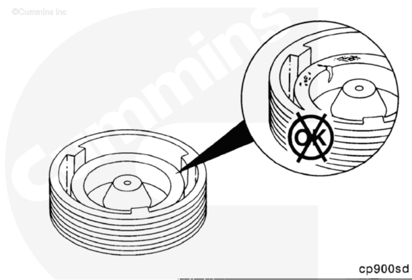
mm		in
4.01	MIN	0.158
4.11	MAX	0.162



Replace the exhaust valve seat if **not** within limits.

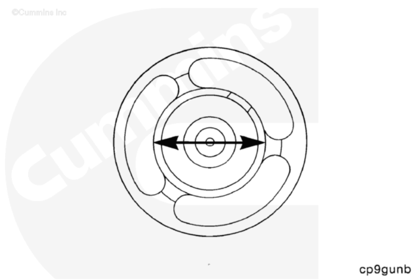
Inlet Valve Cage

Inspect the inlet valve cage for damage and wear.



Measure the valve guide diameter.

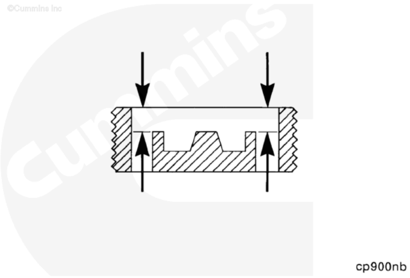
mm		in
25.53	MIN	1.005
25.65	MAX	1.010



Replace the inlet valve cage if **not** within limits.

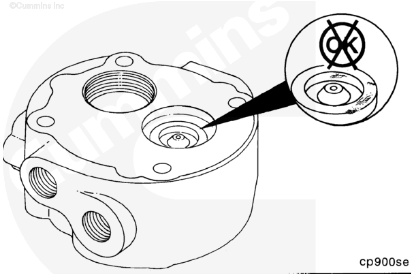
Measure the top of the cage to the valve stop.

mm		in
3.63	MIN	0.143
3.78	MAX	0.149



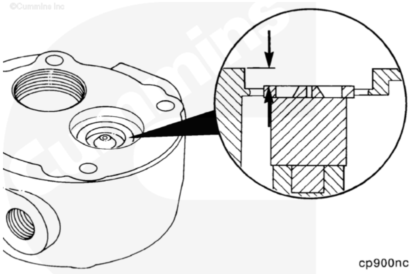
Exhaust Valve Stop

Inspect the exhaust valve stop. Replace the cylinder head assembly if the stop is loose or damaged.



Measure the distance from the valve end of the stop to the face of the cylinder head.

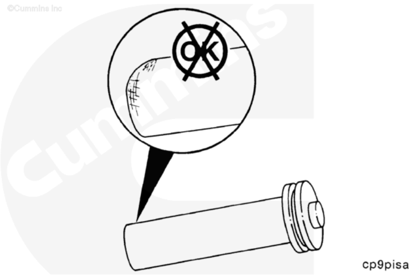
mm		in
4.42	MIN	0.174
4.70	MAX	0.185



Replace the cylinder head if **not** within limits.

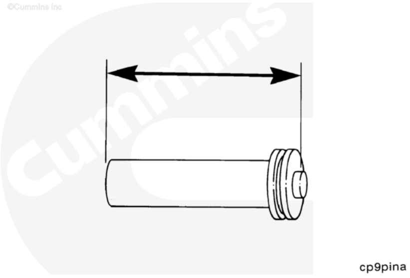
Unloader Pin

Inspect for scoring or pitting.



Measure the length of the pin.

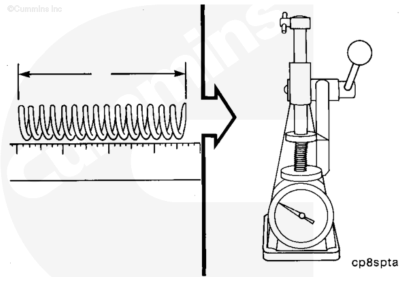
mm		in
40.51	MIN	1.595
40.72	MAX	1.603



Replace the pin if **not** within limits.

Valve Springs

Valve Spring Tester, Part Number 3375182



Note : Cummins Inc. recommends that new springs be installed during rebuild.

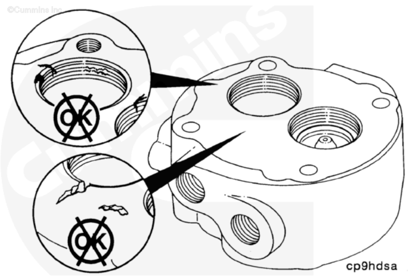
Use valve spring tester, Part Number 3375182, to check the springs.

Replace if **not** within limits in Table 1, shown below.

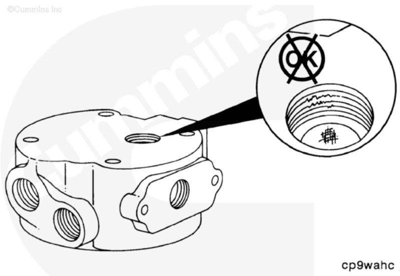
Table 1: Spring Data

Load Required to Compress Spring to Length						
	Spring Length		**Minimum**	**Maximum**		
	mm	in	Kg	lb	Kg	lb
Inlet Valve	5.08	0.20	0.272	0.60	0.340	0.75
Unloader	10.0330	0.395	1.53	3.37	1.90	4.19
Exhaust Valve	5.08	0.20	0.272	0.60	0.340	0.75

Cylinder Head



Inspect and replace if cracks, nicks, gouges, or damaged threads are found.



Inspect the unloader seal bore for scoring or pitting.

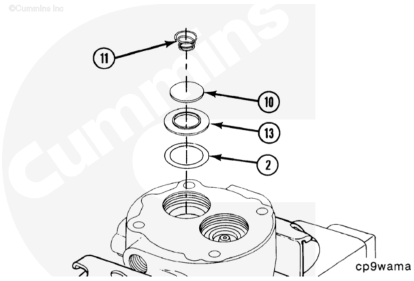
Replace the cylinder head if damaged.

Assemble

Turn the cylinder head bottom side up, and install it in a soft-jawed vise.

Install the following parts.

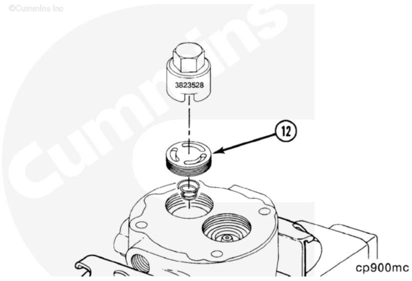
a. New washer (2)
b. Inlet valve seat (13)
c. Inlet valve (10)
d. Inlet valve spring (11).



Air Compressor Seat Socket, Part Number 3823528

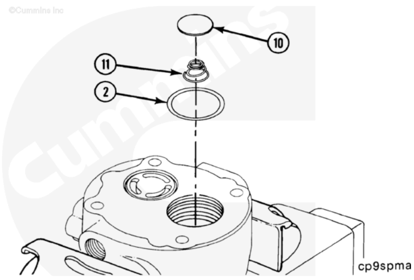
Tighten the cage (12).

Torque Value: 108 n•m [80 ft-lb]



Install the following parts:

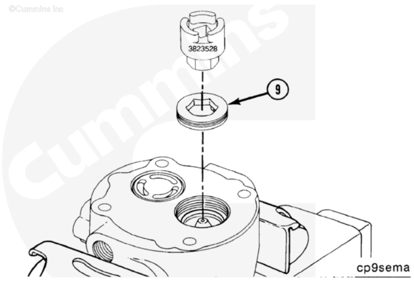
a. New washer (2)
b. Exhaust valve spring (11)
c. Exhaust valve disc (10).



Air Compressor Seat Socket, Part Number 3823528, or 3/4-Inch Allen Wrench

Tighten the seat (9).

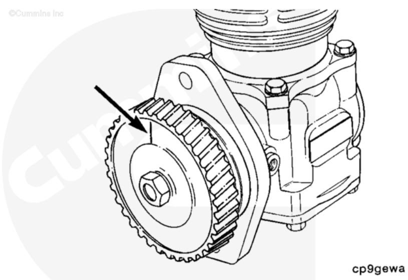
Torque Value: 108 n•m [80 ft-lb]



Prior to installing the cylinder head, locate top dead center of the piston on the air compressor.

Rotate the air compressor so that the piston is at the top.

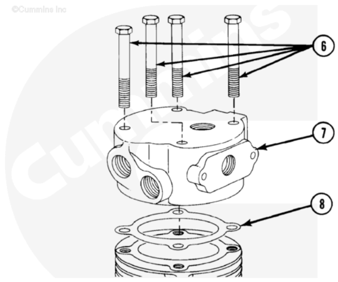
Use ink or Dykem® to mark the air compressor gear face at top dead center (12-o'clock position when viewed from the front).



Note : Top dead center does not have to be exact. The system is tolerant of some misalignment.

Install a new gasket (8) and the cylinder head (7) to the cylinder block, aligning the scribe marks.

Install the four capscrews (6).



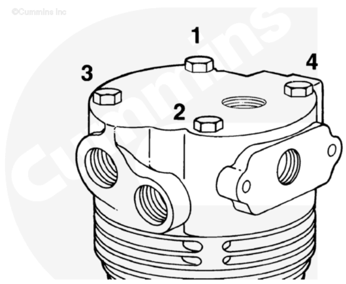
cp9csma

Tighten the capscrews.

Torque Value: 30 n•m [22 ft-lb]

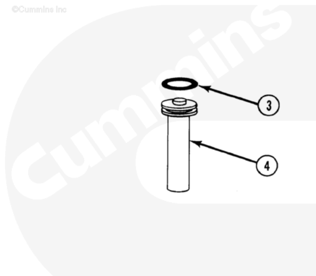
Tighten the capscrews again, in the sequence shown.

Torque Value: 41 n•m [30 ft-lb]



cp9csna

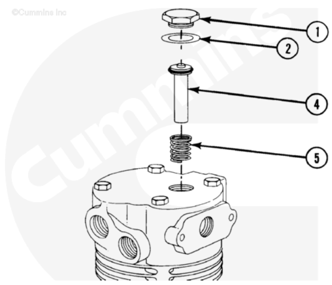
Install a new o-ring (3) on the unloader pin (4).



cp9orma

Install the following parts:

a. Spring (5)
b. Unloader pin (4)
c. New washer (2)
d. Unloader cover (1).



cp9hdma

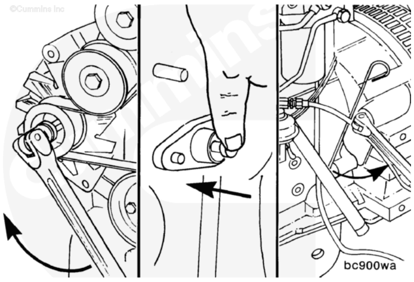
Tighten the cover.

Torque Value: 41 n•m [30 ft-lb]

Install

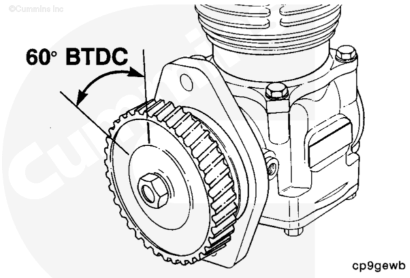
Note : Be sure to disengage the timing pin after locating top dead center .

Locate top dead center for cylinder Number 1 by barring the crankshaft slowly while pushing on the timing pin.



Rotate the compressor top dead center mark to 60 degrees, or 6 teeth on a 36-tooth gear, before top dead center. This is approximately 10-o'clock when viewed from the front of the air compressor.

Note : Holset® air compressors Series SS, QE296 and 338 will have a radial line etched on the gear representing top dead center .

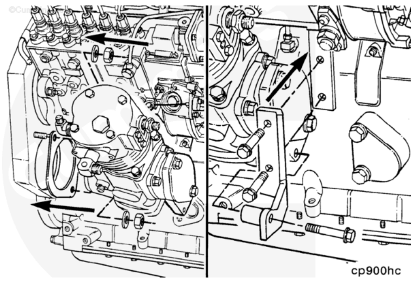


Use a new gasket. Install the air compressor to the gear housing.

Install the air compressor support bracket.

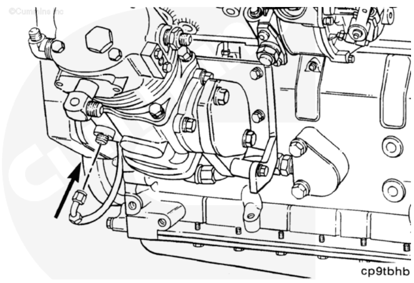
Mounting Nuts	77 n.m	[57 ft-lb]
Support Capscrews	24 n.m	[18 ft-lb]

Note : No timing of gears is necessary.



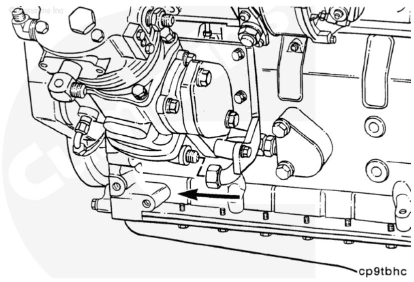
Install the oil supply line.

Torque Value: 15 n•m [133 in-lb]



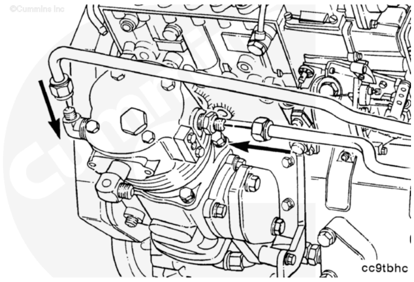
Install the oil drain to the bottom of the compressor.

Torque Value: 24 n•m [18 ft-lb]



⚠ **CAUTION** ⚠

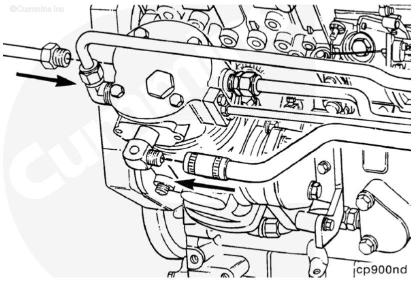
If rubber grommets are used on the coolant or air lines, be sure they are installed carefully to prevent cuts or tears to the grommets, which will cause leaks.



Install the coolant lines.

Torque Value: 24 n•m [18 ft-lb]

Install and tighten the air inlet and outlet connections.



Inlet	5 n.m	[44 in-lb]
*Outlet	24 n.m	[18 ft-lb]

*1/2 NPT fitting in head

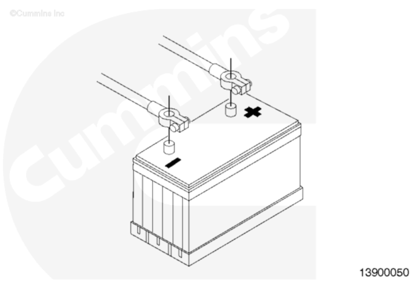
Note : Torque of the discharge line is dependent upon line size and type. Consult vehicle manufacturer for correct torque value.

Finishing Steps

- Fill the engine cooling system (liquid cooled air compressor **only**). Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html).

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries
- Close the wet tank draincock
- Operate the engine to activate the air compressor. With the air compressor pumping between 550 to 690 kPa [80 to 100 psi], use a solution of soapy water to check for air leaks.

Last Modified: 31-Mar-2006